

TARFALA, Sweden

WMO: 020290

Lat: 67.912N

Lon: 18.610E

Elev: 1146

StdP: 88.29

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-19.6	-29.6	0.3	-16.0	-26.9	0.4	-16.5	17.2	-4.3	15.5	-4.1	1.7	330	0.909

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	16.2	10.6	14.4	9.9	12.6	8.8	11.7	15.0	10.5	13.5	9.5	12.0	3.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.9	12.4	9.0	8.2	11.3	8.0	7.7	10.6	36.5	15.0	33.5	13.5	30.8	12.0	15.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
15.2	13.1	11.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	-24.4	11.9	2.7	1.7	-26.3	13.1	-27.8	14.1	-29.3	15.0	-31.2	16.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.7	-10.6	-10.3	-9.4	-5.5	-0.7	4.0	8.5	7.3	2.5	-3.3	-7.0	-8.1
	DBStd	7.79	5.41	4.91	4.48	4.05	3.82	3.46	3.16	3.23	3.66	4.17	4.05	4.70
	HDD10.0	4663	639	567	602	466	331	182	69	97	224	413	509	562
	HDD18.3	7669	897	801	861	716	589	431	306	342	474	672	759	821
	CDD10.0	36	0	0	0	0	0	1	21	13	1	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.9	4.0	4.3	4.2	3.7	3.8	3.7	3.3	3.2	3.9	4.0	3.6	4.9
Precipitation	PrecAvg	826	78	66	64	42	47	60	80	75	80	84	67	83
	PrecMax	933	174	129	138	91	76	115	151	131	140	135	131	163
	PrecMin	666	9	9	21	18	17	12	26	34	26	32	18	31
	PrecStd	78	40	26	31	18	16	22	30	23	29	27	31	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	1.9	3.4	7.0	10.0	14.1	19.8	17.5	12.8	8.6	4.8	3.2
		MCWB	-0.6	-1.5	-1.3	3.2	5.6	8.6	12.9	11.1	6.7	5.3	1.3	0.5
	2%	DB	0.2	-0.1	0.3	4.3	8.1	12.2	17.4	15.6	10.4	5.6	2.2	1.3
		MCWB	-1.6	-1.7	-2.1	1.2	4.1	7.3	11.6	10.4	6.6	2.7	0.5	-0.5
	5%	DB	-0.9	-1.4	-1.1	2.3	6.7	10.7	15.4	13.8	8.9	3.9	0.6	0.0
		MCWB	-2.5	-3.1	-2.9	-0.1	3.4	6.6	10.5	9.8	5.9	2.0	-1.0	-1.4
	10%	DB	-2.5	-2.7	-2.7	0.5	4.8	9.1	13.5	12.2	7.5	2.8	-0.8	-1.8
		MCWB	-4.3	-4.2	-4.2	-1.3	2.2	5.8	9.5	9.1	5.2	1.2	-2.3	-3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	-0.1	-0.3	3.2	5.9	9.9	13.7	13.4	9.0	6.1	2.0	1.2
		MCDB	1.2	0.6	1.7	6.3	9.3	11.6	17.5	16.0	10.5	8.4	4.2	2.5
	2%	WB	-1.3	-1.9	-1.9	1.7	4.6	8.6	12.2	11.5	7.6	3.6	0.5	-0.2
		MCDB	0.0	-0.4	0.1	3.9	7.5	10.8	16.1	14.1	9.2	4.8	2.3	1.1
	5%	WB	-2.5	-3.0	-2.9	0.3	3.4	7.3	11.1	10.3	6.4	2.4	-0.9	-1.5
		MCDB	-1.0	-1.3	-1.1	2.0	5.9	9.9	14.5	13.2	8.0	3.6	0.5	0.1
	10%	WB	-4.2	-4.2	-4.2	-1.1	2.3	6.2	10.0	9.2	5.5	1.3	-2.3	-3.3
		MCDB	-2.7	-2.6	-2.7	0.3	4.4	8.7	12.9	11.6	7.1	2.6	-1.0	-2.0
Mean Daily Temperature Range	5% DB	MDBR	5.8	5.6	5.2	5.4	4.7	5.0	5.5	4.7	4.1	4.5	5.0	5.4
		MCDBR	6.4	6.5	6.1	6.2	6.0	7.0	7.9	7.0	5.8	5.4	5.9	6.5
	5% WB	MCWBR	5.7	5.8	5.2	4.8	4.5	4.3	4.4	4.3	4.2	4.3	5.2	5.9
		MCWBR	6.1	6.3	5.6	5.5	5.7	6.2	6.9	6.1	4.8	5.3	5.8	6.1
Clear-Sky Solar Irradiance	taub	taub	0.153	0.210	0.248	0.271	0.287	0.293	0.300	0.289	0.262	0.225	0.157	N/A
		taud	1.921	2.152	2.242	2.279	2.379	2.500	2.549	2.560	2.589	2.295	1.954	N/A
	Edn at Noon	Edn at Noon	168	638	812	889	912	916	898	873	819	614	143	0
		Edh at Noon	8	43	73	95	97	89	82	73	54	37	7	0
All-Sky Solar Radiation	RadAvg	RadAvg	0.04	0.46	1.88	3.83	4.96	4.96	4.57	3.33	1.77	0.59	0.10	0.00
	RadStd	RadStd	0.00	0.05	0.11	0.18	0.38	0.41	0.45	0.31	0.18	0.08	0.01	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page